

Term and Definition Cards

126 cards selected from the practice midterm, the assigned problem sets, the textbook's key-concept lists and the lecture slides. Cover the right column and define each term out loud.

★ = tested directly on the practice midterm (65 cards). Learn those first.

PE Q14 practice exam question
 PS2 · Ch 3 #7 problem set or textbook problem
 Ch 6 key concept textbook key-concept list
 L3 lecture slides
 BG standard theory not in the files

Week 1: Data and Measurement

1	Recession	A significant temporary decline in economic activity throughout the economy (output, employment, trade, productivity), generally lasting at least two quarters of a year. L1
2	Gross domestic product (GDP)	The market value of all final goods and services produced in an economy in a period. It equals total expenditure, total income, and total value added (the circular flow). L2 Ch 2
3	Expenditure identity	$Y = C + I + G + NX$ Output equals consumption plus investment plus government purchases plus net exports. L2 Ch 2
4	Consumption (C)	Spending by households on goods and services. The largest component of GDP, roughly two-thirds. L2
5	★ Investment (I)	Goods purchased by individuals or businesses for future use, such as machines, buildings, and inventories. PE Q19 Ch 2 #4
6	Government purchases (G)	Goods and services bought by government (for example, computers for a public school). Transfer payments are not included. Ch 2 #4 BG
7	Net exports (NX)	Exports minus imports. A US good sold to a buyer abroad counts in NX. Ch 2 #4
8	Stock vs flow	A flow is measured per period of time (GDP, investment, monthly job gains). A stock is measured at a point in time (capital stock, number employed). Bathtub: the faucet is the flow, the water in the tub is the stock. L2

9	Value added	The value of a firm's output minus the value of the intermediate goods it buys. GDP is the sum of value added across all firms. Ch 2 BG
10	Nominal GDP	Output valued at current prices: $\text{Nominal GDP}_t = \sum_i P_{i,t} Q_{i,t}$ Ch 2 #7
11	Real GDP	Output valued at base-year prices, so it changes only when quantities change: $\text{Real GDP}_t = \sum_i P_{i,\text{base}} Q_{i,t}$ Ch 2 #7
12	★ GDP deflator	$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$ Prices of everything produced domestically , with a basket that changes with current output (a Paasche index). Excludes imports. PE Q1 Ch 2 #7
13	★ Consumer price index (CPI)	$\text{CPI}_t = \frac{\sum_i P_{i,t} Q_{i,\text{base}}}{\sum_i P_{i,\text{base}} Q_{i,\text{base}}} \times 100$ Price of a fixed basket bought by a typical consumer (a Laspeyres index). Includes imports consumers buy. PE Q1 Ch 2 #7
14	★ CPI vs GDP deflator (the tested difference)	The CPI includes imported goods; the deflator includes only domestically produced goods. The CPI uses a fixed basket; the deflator's basket changes. Both cover goods <i>and</i> services. PE Q1
15	Laspeyres vs Paasche bias	A fixed-basket (Laspeyres) index such as the CPI overstates increases in the cost of living because it ignores substitution. A changing-basket (Paasche) index such as the deflator understates them. Ch 2 #7-8
16	Inflation rate	$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}$ The percentage change in a price index. Ch 2
17	Labor force	$L = E + U$ Everyone employed plus everyone unemployed. Ch 2 #9
18	Unemployed	Without a job, available to work, and actively searching. Ch 2 #9

19	Not in the labor force	Adults who are neither employed nor searching: retirees, students, homemakers, and discouraged workers. Ch 2 #9
20	★ Discouraged worker	Someone who wants a job but has stopped searching. Counted as not in the labor force , so not as unemployed. PE Q3 Ch 2 #9
21	Unemployment rate	$u = \frac{U}{L} = \frac{U}{E + U}$ Ch 2 #9
22	Labor force participation rate	$\text{LFPR} = \frac{L}{\text{adult population}}$ Ch 2 #9
23	Employment-population ratio	$\frac{E}{\text{adult population}}$ Not affected by people moving between unemployment and out of the labor force. L2
24	Household survey vs establishment survey	The household survey counts <i>people</i> employed (basis of the unemployment rate). The establishment (payroll) survey counts <i>jobs</i> : a person with two jobs counts twice, and the self-employed are not counted. Ch 2 #9c
25	GDP and well-being	GDP omits leisure, home production, health, and the environment, and can rise with activity that lowers well-being. It is still useful because it correlates with much of what people value. Ch 2 #10–11

Week 2: National Income, Distribution and Loanable Funds

26	Production function	$Y = F(K, L)$ Output depends on capital, labor, and technology. In the long-run model, factor supplies are fixed, so output $\bar{Y}$ is fixed. Ch 3 L3
27	★ Constant returns to scale	$F(zK, zL) = z F(K, L)$ Scaling all inputs by z scales output by z . Ch 3 #3 PE Q13c
28	★ Cobb-Douglas production function	$Y = A K^{\alpha} L^{1-\alpha}$ α is capital's share of income; A is productivity (technology). PE P2 Q3 Ch 3

29	★ Total factor productivity (A)	The technology term in the production function. Higher A raises output, the real wage, and the real rental rate together. PE P2 Q3b Ch 3 #1
30	Marginal product of labor (MPL)	Extra output from one more unit of labor: $\text{MPL} = \frac{\partial F}{\partial L}$ It falls as L rises (diminishing marginal product). Ch 3
31	Marginal product of capital (MPK)	$\text{MPK} = \frac{\partial F}{\partial K}$ Falls as K rises. Ch 3
32	★ Firm's profit	$\text{Profit} = PY - WL - RK$ W is the nominal wage, R the nominal rental rate of capital. PE P2 Q3a
33	★ Labor demand condition	Firms hire until the marginal product of labor equals the real wage: $\text{MPL} = \frac{W}{P}$ PE P2 Q3a PS2
34	★ Capital demand condition	Firms rent capital until $\text{MPK} = \frac{R}{P}$ the real rental price of capital. PE P2 Q3b PS2
35	Real wage	The nominal wage divided by the price level, W/P . In equilibrium it equals MPL. Ch 3 #2
36	★ Cobb-Douglas marginal products	$\text{MPL} = (1 - \alpha) \frac{Y}{L} \quad \text{MPK} = \alpha \frac{Y}{K}$ PE P2 Q3 Ch 3 #4
37	★ Labor share of income	$\frac{WL}{PY} = \frac{\text{MPL} \cdot L}{Y} = 1 - \alpha$ Constant under Cobb-Douglas, even when K or L change. PE Q13a PE P2 Q3b
38	★ Capital share of income	$\frac{RK}{PY} = \frac{\text{MPK} \cdot K}{Y} = \alpha$ PE P2 Q3b
39	★ Backing out α and A	Use the labor share, then the production function: $1 - \alpha = \frac{\text{MPL} \cdot L}{Y}, \quad A = \frac{Y}{K^\alpha L^{1-\alpha}}$ PE P2 Q3b

40	Zero economic profit	With competition and constant returns to scale, paying each factor its marginal product uses up all output: $Y = MPL \cdot L + MPK \cdot K$ Ch 3
41	Neoclassical theory of distribution	Each factor is paid its marginal product. More labor lowers the real wage and raises the rental rate; less capital does the same; better technology raises both. Ch 3 #1 PS2
42	★ Real wage and labor productivity	$\frac{W}{P} = (1 - \alpha) \frac{Y}{L}$ Real wages are proportional to average labor productivity under competition and constant returns. PE Q13c L3
43	★ Declining labor share: explanations	(1) Production technology : offshoring and automation (capital displacing workers). (2) Rising market power : higher markups ($P > MC$), superstar firms, concentration that lowers wages relative to productivity. PE Q13d L3
44	Markup	Price above marginal cost ($P > MC$). Rising markups mean firms keep more of output, lowering the labor share. L3
45	★ Wage compression after the pandemic	Since the pandemic, wage growth has been fastest for low-income workers (10th percentile). PE Q13b L3
46	Disposable income	$Y - T$ Income after taxes. Ch 3
47	★ Consumption function	$C = C(Y - T)$ Consumption depends on disposable income. PE Q5 PE Q18
48	★ Marginal propensity to consume (MPC)	The extra consumption from an extra dollar of disposable income: $MPC = \frac{\partial C(Y - T)}{\partial (Y - T)}, \quad 0 < MPC < 1$ PE Q18
49	★ Investment function	$I = I(r)$ Investment falls as the real interest rate rises (borrowing costs more). PE Q5 Ch 4
50	★ Goods-market equilibrium	$\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}$ The real interest rate r adjusts so demand equals the fixed supply of output. PE Q5 PE P2 Q3c

51	Private saving	$S_{\text{private}} = Y - T - C$ Ch 4 #1, #3
52	Public saving	$S_{\text{public}} = T - G$ Negative means a budget deficit. Ch 4 #3
53	National saving	$S = Y - C - G = (Y - T - C) + (T - G)$ Ch 4 #3 PS3
54	★ Loanable funds market	Saving (supply) equals investment (demand): $S = I(r)$ Graph: r on the vertical axis; saving is a vertical line; investment demand slopes down . PE Q5 Ch 4
55	★ Crowding out	Higher government spending lowers public saving, shifts the saving line left, raises r , and reduces investment one-for-one (Textbook Figure 3-9). PE Q5 L3
56	Tax change and saving	A tax increase ΔT raises public saving by ΔT and lowers private saving by $\text{MPC} \cdot \Delta T$: $\Delta S = (1 - \text{MPC}) \Delta T$ National saving rises, r falls, investment rises. Ch 4 #1
57	Balanced-budget increase	Raising G and T by the same x : $\Delta S = -(1 - \text{MPC}) x$ so r rises and investment falls by $(1 - \text{MPC})x$. PS2 (Ch 4 #4)
58	★ Exogenous variable	A variable known beforehand and fed into the model (for example K, L, G, T). PE Q14
59	★ Endogenous variable	A variable of interest that comes out of the model (for example r, C, I). PE Q14

Week 3: Money, Inflation and Monetary Policy

60	Inflation	A sustained increase in the overall price level. Ch 6 key concept
61	Hyperinflation	Extremely high inflation (Mankiw: above 50% per month). Starts when governments finance large deficits by printing money; ends with fiscal reform that removes the need for seigniorage. Example: Germany 1922–23. Ch 6 key concept L5 BG

62	Money	The stock of assets used for transactions. Functions: store of value, unit of account, medium of exchange. BG (Lecture 6 not in files)
63	Quantity equation	$M \times V = P \times Y$ Money times velocity equals nominal GDP. Ch 6 key concept PS3
64	Transactions vs income velocity	Transactions velocity uses total transactions: $MV = PT$. Income velocity uses output: $MV = PY$. The course uses income velocity. Ch 6 key concept
65	★ Velocity of money	How many times a dollar changes hands per period: $V = \frac{PY}{M}$ PE Q11 Ch 6 #3
66	Real money balances	$\frac{M}{P}$ The purchasing power of the money stock. Ch 6 key concept
67	Money demand function	$\left(\frac{M}{P}\right)^d = kY \Rightarrow V = \frac{1}{k}$ More generally $(M/P)^d = L(i, Y)$: demand falls as i rises, so velocity rises with i . Ch 6 #2-4
68	★ Quantity theory of money	With velocity constant and output fixed by factors, the money supply determines nominal GDP and the price level: P is proportional to M . PE Q15 Ch 6 key concept
69	Quantity theory in growth rates	$\% \Delta M + \% \Delta V = \% \Delta P + \% \Delta Y$ With constant velocity: $\pi = \% \Delta M - \% \Delta Y$ Ch 6 #1 PS3
70	★ Seigniorage	Revenue the government raises by printing money. PE Q15 Ch 6 key concept
71	★ Inflation tax	Printing money raises inflation (quantity theory), so money held by the public loses value while the government gains revenue. It works like a tax on people who hold money . PE Q15
72	Nominal vs real interest rate	The nominal rate i is what the bank pays. The real rate r is the increase in purchasing power: $r = i - \pi$. Ch 6 key concept

73	Fisher equation	$i = r + \pi$
		PS3 Ch 6 #1
74	Fisher effect	A one-point rise in expected inflation raises the nominal interest rate one point and leaves the real rate unchanged (one-for-one).
		Ch 6 #3d Ch 6 key concept L5
75	Ex ante vs ex post real interest rate	$r^{\text{ex ante}} = i - E[\pi] \quad r^{\text{ex post}} = i - \pi$ Ex ante uses expected inflation; ex post uses actual.
		Ch 6 key concept
76	Shoeleather costs	Costs of holding less money because of inflation, such as more trips to the bank.
		Ch 6 key concept Ch 6 #7
77	Menu costs	Costs of changing prices, such as reprinting catalogs and menus.
		Ch 6 key concept Ch 6 #7
78	Other costs of expected inflation	Relative-price variability (prices change at different times, so resources are misallocated), tax distortions (nominal gains are taxed), and inconvenience of a changing unit of account.
		Ch 6 #5, #7
79	Cost of unexpected inflation	Arbitrary redistribution between debtors and creditors. Lenders with fixed nominal payments (for example, an annuity) lose; borrowers gain.
		Ch 6 #7b
80	Possible benefit of inflation	Lets real wages fall without nominal wage cuts, and allows negative real interest rates.
		Ch 6
81	★ Real vs nominal variables	Nominal variables are measured in money (prices, nominal wages, i). Real variables are measured in physical units (output, real wage, r).
		Ch 6 key concept PE Q20
82	★ Classical dichotomy	The theoretical separation of real and nominal variables: nominal variables do not affect real variables . (The reverse statement is false.)
		PE Q20 Ch 6 key concept
83	Monetary neutrality	Changes in the money supply do not affect real variables in the long run.
		Ch 6 key concept
84	★ Monetary base	Currency in circulation plus bank reserves. Not the sum of all currency and deposits.
		PE Q16c
85	★ M1	Currency plus checkable deposits (and, since a 2020 Fed regulation change, savings deposits, which created an artificial jump in measured M1).
		PE Q16b PS2

86	★ Open market operations	The Fed adjusts the money supply by buying or selling assets such as short-term government debt. PE Q16d
87	★ Forward guidance	The Federal Reserve providing information about its expected future policy actions. It helps keep the economy stable by anchoring public expectations about future inflation. PE Q2
88	Federal funds rate	The overnight interest rate the FOMC targets. L5
89	★ Causes of the recent inflation	Supply: COVID disrupted supply chains and the workforce. Demand: monetary and fiscal stimulus boosted spending. PE Q10
90	Classical combined model	The real side sets r from $S = I(r)$; money growth sets π ; then $i = r + \pi$. Higher G raises r and i but not inflation. PS3

Week 4: Unemployment and Labor-Market Flows

91	Natural rate of unemployment	The average rate around which actual unemployment fluctuates; in the flows model, the steady-state rate. L7
92	★ Job separation rate (s)	The fraction of employed workers who lose their jobs each period. PE P2 Q2
93	★ Job finding rate (f)	The fraction of unemployed workers who find jobs each period. PE P2 Q2
94	★ Unemployment dynamics	$U_{t+1} = (1 - f) U_t + s E_t$ PE P2 Q2
95	★ Employment dynamics	$E_{t+1} = (1 - s) E_t + f U_t$ PE P2 Q2
96	★ Steady-state condition	Flows into unemployment equal flows out: $sE = fU$ PE P2 Q2b
97	★ Steady-state unemployment rate	Substitute $E = L - U$ into $sE = fU$: $\frac{U}{L} = \frac{s}{s + f}$ Example: $s = 0.02$, $f = 0.28 \Rightarrow u = 6.67\%$. PE P2 Q2

98	★ Frictional unemployment	Unemployment from the time it takes to match workers and jobs (job search). Updating a resume or holding out for a good skill match is frictional. PE Q8
99	★ Structural unemployment	Unemployment from wages held above the market-clearing level, so labor supplied exceeds labor demanded. Reasons given in class: unions and a minimum wage above equilibrium. PE Q8
100	★ Wage rigidity	Failure of wages to fall to the level where supply equals demand; the source of structural unemployment. PE Q8 BG
101	★ Unions and unemployment	Unions can push members' wages above the going rate in non-union workplaces, reducing jobs (structural unemployment). PE Q8b
102	★ Minimum wage and unemployment	A minimum wage well above equilibrium creates structural unemployment. PE Q8c L3
103	Efficiency wages	Firms pay above market wages to raise productivity (less turnover, more effort). Another source of wage rigidity; not in the practice key, so confirm in your notes. BG
104	★ Unemployment insurance (UI)	A program funded by taxes on employers (and employees) that replaces part of an unemployed worker's income (usually 30–50%) for a limited time (usually up to 26 weeks) while they search. PE Q12
105	★ Why the unemployment rate is imperfect	It ignores discouraged workers outside the labor force. If unemployed people give up and leave the labor force, the rate falls even though conditions did not improve. PE Q3

Week 5: Growth and the Solow Model

106	★ Income per worker as a measure of well-being	Justified because it is correlated with many other things we care about , and it is easy to compute consistently for many countries. PE Q7
107	Solow growth model	A model of how saving, depreciation, and (in extensions) population growth and technology determine capital per worker and income per worker over time. Ch 9 key concept
108	★ Capital per worker and output per worker	$k = \frac{K}{L}, \quad y = \frac{Y}{L} = f(k) = k^\alpha$ PS5 PE P2 Q1

109	★ Investment and consumption per worker	$i = s f(k), \quad c = (1 - s) f(k)$ s is the saving rate. PS5 PE P2 Q1
110	★ Depreciation (δ)	The fraction of the capital stock that wears out each period; depreciation per worker is δk. PS5 PE P2 Q1
111	★ Capital accumulation	$K_{t+1} = (1 - \delta)K_t + I_t \quad \Delta k = s f(k) - \delta k$ PE P2 Q1 PS5
112	Steady state	The capital per worker at which investment just covers depreciation, so k stops changing: $s f(k^*) = \delta k^*$ Ch 9 key concept PS5
113	Steady-state capital per worker	For $y = k^\alpha$: $k^* = \left(\frac{s}{\delta}\right)^{\frac{1}{1-\alpha}}, \quad y^* = \left(\frac{s}{\delta}\right)^{\frac{\alpha}{1-\alpha}}, \quad c^* = (1 - s) y^*$ PS5
114	★ Solow diagram	Capital per worker k on the horizontal axis; concave curve $s f(k)$; straight line δk from the origin; they cross at k^*. Left of k^*, k rises; right of k^*, k falls. Label every line and intersection. PE P2 Q1b
115	★ Saving rate and growth	A higher saving rate raises the steady-state level of income per worker, with only temporary faster growth. It does not raise the long-run growth rate. PE Q17 Ch 9 #5
116	★ Golden rule level of capital	The steady-state capital per worker that maximizes consumption per worker. Ch 9 key concept PS5 PE P2 Q1a
117	★ Golden rule condition	Maximize $c^* = f(k^*) - \delta k^*$: $f'(k_{\text{gold}}) = \delta \quad (\text{MPK} = \delta)$ PE P2 Q1a Ch 9 #2d
118	★ Golden rule capital (Cobb-Douglas)	$\alpha k^{\alpha-1} = \delta \Rightarrow k_{\text{gold}} = \left(\frac{\alpha}{\delta}\right)^{\frac{1}{1-\alpha}}$ PE P2 Q1a PS5
119	★ Golden rule saving rate	Plug k_{gold} into $s f(k) = \delta k$: $s_{\text{gold}} = \alpha$ Show this derivation on the exam; stating it alone loses points. PE P2 Q1a PS5
120	Below vs above the golden rule	Below ($k^* < k_{\text{gold}}$): raising saving lowers consumption now and raises it later. Above: cutting saving raises consumption at every date. Ch 9 summary PS5

121	★ Fall in the labor force	K unchanged, L falls, so $k = K/L$ jumps above k^*. Income per worker jumps up, then declines (negative growth) back to the same steady state. Total output falls. PE P2 Q1b PS5 (Ch 9 #3)
122	★ Consumption after a labor-force fall	With s unchanged, $c = (1 - s)y$ follows income: jumps up, then declines back to its original steady-state level. PE P2 Q1c
123	Population growth in the Solow model	With population growth n: $\Delta k = s f(k) - (\delta + n) k$ Faster population growth lowers steady-state capital and income per worker. BG (Lecture 9 slides empty)
124	★ Population growth and ideas	Declining population growth may slow innovation, because fewer people are generating new ideas. (True on the practice exam.) PE Q4
125	★ Why China's growth slowed	Unsustainable policy boosts; accumulated inefficiencies after long fast growth; recessions abroad hurt export demand; nearing its steady state; pivot to consumption-led growth (lower saving). PE Q6 L8
126	Catch-up (convergence) growth	Economies far below their steady state grow fast as capital accumulates, then slow. Example: postwar Japan and Germany. L8 Ch 9